

Packet Capture and Analysis Report Using Wireshark

Experiment Title: Network Traffic Analysis Using Wireshark

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Tool Used: Wireshark

Operating System: Windows 11, Kali linux

1. Objective:-

The objective of this task is to capture live network traffic using **Wireshark**, analyze different network protocols, and understand packet-level communication by inspecting real-time data.

2. Tool Description:-

Wireshark is a network protocol analyzer that allows users to capture and interactively browse the traffic running on a computer network. It is widely used for network troubleshooting, analysis, and security auditing.

3. Methodology / Procedure:-

The following steps were performed during the experiment:

1. Installed **Wireshark** on the system.
2. Opened Wireshark and selected the **active network interface** (Wi-Fi/Ethernet).
3. Started packet capture.
4. Generated network traffic by:
 - Browsing a website (e.g., google.com)
 - OR pinging a server.
5. Captured packets for approximately **1 minute**.
6. Stopped the packet capture.
7. Applied protocol filters such as:
 - http
 - dns
 - tcp
8. Identified different protocols from the captured traffic.
9. Exported the capture file in **.pcap** format.
10. Analyzed packet details and summarized the findings.

4. Observed Protocols

At least three different protocols were identified during the capture:

Protocol	Description
TCP	Ensures reliable, ordered delivery of packets
DNS	Resolves domain names to IP addresses
HTTP / HTTPS	Used for web communication

5. Packet Analysis Details

5.1 TCP (Transmission Control Protocol)

- Used for reliable communication between client and server.
- Observed **3-way handshake** (SYN, SYN-ACK, ACK).
- Commonly used for web browsing and secure connections.

5.2 DNS (Domain Name System)

- DNS query packets were observed when visiting websites.
- Translates domain names (e.g., google.com) into IP addresses.
- Uses UDP port 53.

5.3 HTTP / HTTPS

- HTTP packets show request and response communication.
- HTTPS traffic is encrypted and uses TCP port 443.
- Website access generated multiple HTTP/HTTPS packets.

6. Findings Summary

- Live network traffic was successfully captured.
- Multiple protocols were observed simultaneously.
- DNS packets appeared before web traffic, indicating name resolution.
- TCP packets ensured reliable data transfer.
- Encrypted HTTPS traffic protects sensitive information.

7. Exported Capture File

- The captured packets were exported as a **.pcap file** for future analysis and reporting.
- Wireshark interface with active capture
- Applied protocol filters
- Packet details pane



network_capture.pcap
png

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8. Conclusion

This experiment provided hands-on experience with **packet capturing and protocol analysis** using Wireshark. It helped in understanding how different protocols function together during normal network communication and improved practical knowledge of network monitoring and cybersecurity fundamentals.

